

Massively Multilingual Joint Segmentation and Glossing

大规模多语言联合分词与词形标注

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Abstract

摘要

Automated interlinear gloss prediction with neural networks is a promising approach to accelerate language documentation efforts. However, while state-of-the-art models like GLOSSLM (Ginn et al., 2024b) achieve high scores on glossing benchmarks, user studies with linguists have found critical barriers to the usefulness of such models in real-world scenarios (Rice et al., 2025). In particular, existing models typically generate morpheme-level glosses but assign them to whole words without predicting the actual morpheme boundaries, making the predictions less interpretable and thus untrustworthy to human annotators.

使用神经网络进行自动的线性注释预测是一种有前景的方法，可以加速语言记录工作。然而，尽管像 GLOSSLM (Ginn 等, 2024b) 这样的最先进的模型在注释基准测试中取得了高分，但与语言学家进行的用户研究发现，这些模型在现实场景中的实用性存在关键障碍 (Rice 等, 2025)。特别是，现有模型通常生成词素级别的注释，但将其分配给整个单词，而没有预测实际的词素边界，这使得预测结果更难以解释，因此对人类标注者来说缺乏可信度。

We conduct the first study on neural models that jointly predict interlinear glosses and the corresponding morphological segmentation from raw text. We run experiments to determine the optimal way to train models that balance segmentation and glossing accuracy, as well as the alignment between the two tasks. We extend the training corpus of GLOSSLM and pretrain POLYGLOSS, a family of seq2seq multilingual models for joint segmentation and glossing that outperforms GLOSSLM on glossing and beats various open-source LLMs on segmentation, glossing, and alignment. In addition, we demonstrate that POLYGLOSS can be quickly adapted to a new dataset via low-rank adaptation.

我们首次研究了神经模型，该模型能够从原始文本中联合预测带注释的词形分析 (interlinear glosses) 和相应的词形分析分割 (morphological segmentation)。我们进行了实验，以确定训练模型的最佳方式，使模型在分割和注释准确性之间取得平衡，并且两个任务之间的对齐也得到优化。我们扩展了 GLOSSLM 的训练语料库，并预训练了 POLYGLOSS，这是一个用于联合词形分析和注释的 seq2seq 多语言模型系列，其在注释任务上优于 GLOSSLM，并在词形分析、注释和对齐任务上优于各种开源大型语言模型 (LLMs)。此外，我们还展示了 POLYGLOSS 可以通过低秩适应 (low-rank adaptation) 快速适应到新的数据集上。

1 Introduction

1 引言

Nearly half of the world’s 7,000 languages face extinction. For many speakers and linguists of these languages, language documentation has become an urgent goal. Documentation projects commonly involve the creation of interlinear glossed text (IGT), a dense annotation format combining morphological segmentation, tagging, and translation (Figure 1). Due to its structured

format and common usage among linguists, IGT has proven useful for linguistic analysis (Bender et al., 2013; Zama-

世界上大约有 7000 种语言，近一半的语言面临灭绝的威胁。对于这些语言的使用者和语言学家来说，语言记录已成为一项紧迫的目标。语言记录项目通常包括创建一种密集的注释格式——交互线性注释文本 (IGT)，它结合了形态学分词、标记和翻译 (图 1)。由于其结构化的格式以及在语言学家中的普遍使用，IGT 已被证明对语言分析非常有用 (Bender 等, 2013; Zama-

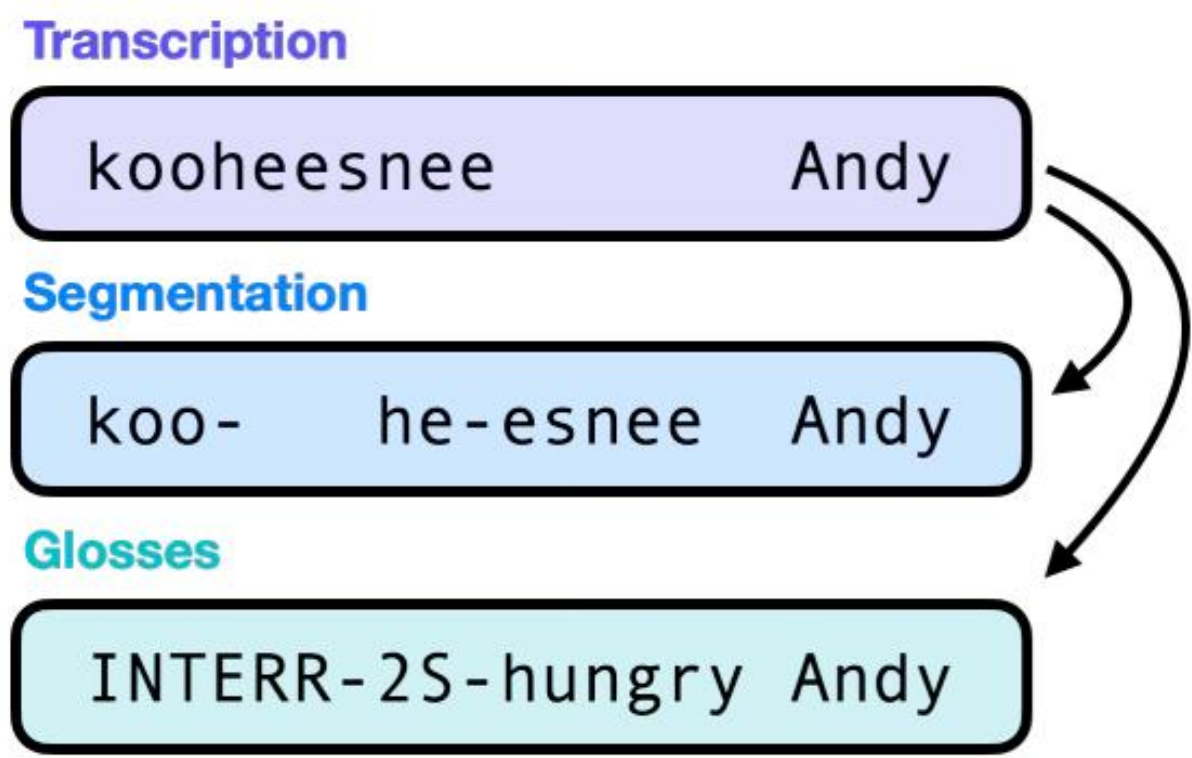


Figure 1: images/449b025535cf60cd6323ef148f470c0eee41a1781d89bbe51146f2baa74d7a44.jpg

Figure 1: An interlinear glossed text example, showing the Arapaho for “Are you hungry, Andy?”. Our model predicts the segmentation and gloss line from the transcribed text.

图 1：一个带注音的平行文本示例，展示了“你饿吗，安迪？”的阿拉帕霍语。我们的模型可以从转录文本中预测分词和注音行。

raeva, 2016; Moeller et al., 2020), language pedagogy (Alast and Baleghizadeh, 2024; Bonilla Carvajal, 2025), and development of language technology such as taggers (Georgi, 2016), searchable text databases (Blokland et al., 2019; Rijhwani et al., 2023), educational tools (Uiibo et al., 2017; Chaudhary et al., 2023), and machine translation systems (Zhou et al., 2020; Ramos et al., 2025).

(raeva, 2016; Moeller et al., 2020), 语言教学 (Alast 和 Baleghizadeh, 2024; Bonilla Carvajal, 2025), 以及语言技术的发展，例如标注工具 (Georgi, 2016)、可搜索文本数据库 (Blokland 等, 2019; Rijhwani 等, 2023)、教育工具 (Uiibo 等, 2017; Chaudhary 等, 2023) 和机器翻译系统 (Zhou 等, 2020; Ramos 等, 2025)。

Creating IGT is expensive, and a number of studies have proposed methods to automate IGT production with statistical and neural methods (McMillan-Major, 2020a; Zhao et al., 2020; Ginn et al., 2024a). In all of these studies, including the 2023 SIGMORPHON shared task (Ginn et al., 2023), the task is formulated as predicting the gloss line from the transcription or segmentation line. The former is more difficult (but also more useful), as it requires the model to infer morphological segmentation in addition to predicting glosses, and has been the primary focus of recent work.

创建 IGT 是昂贵的，许多研究提出了利用统计和神经方法自动化 IGT 生产的方法（McMillan-Major, 2020a; Zhao 等, 2020; Ginn 等, 2024a）。在所有这些研究中，包括 2023 年 SIGMORPHON 共享任务（Ginn 等, 2023），任务都被表述为从转录线或分段线预测词形标注线。前者更为困难（但也更有用），因为它要求模型在预测词形标注的同时推断形态学分段，这也是近期工作的主要关注点。

Though state-of-the-art glossing models such as GLOSSLM (Ginn et al., 2024b) have achieved high accuracy across many languages, Rice et al. (2025) discovered several issues when using these models in a realistic documentation scenario:

尽管像 GLOSSLM（Ginn 等, 2024b）这样的最先进的标注模型在许多语言中取得了高准确率，但 Rice 等（2025）在将这些模型应用于现实文档场景时发现了一些问题：

1. First, documentary linguists typically perform explicit morphological segmentation before glossing each morpheme, so a model that produces glosses directly—without exposing the implicit segmentation—is confusing, less interpretable, and difficult to trust.
2. Second, the model produced very inaccurate glosses in two of the three languages studied, with the participants agreeing that correcting the predicted outputs would be more difficult than annotating from scratch, or using a simpler lookup-based method.
3. Third, the model often predicted gloss labels which were unfamiliar or unlike the glossing conventions preferred by the participants, and the existing system provided no way to adapt its labels to the preferred conventions.
4. 首先，文档语言学家通常在对每个语素进行注释之前，会先明确地进行形态学切分，因此一个直接生成注释而不暴露隐含切分过程的模型会令人困惑，解释性较差，也难以让人信任。
5. 其次，在所研究的三种语言中，该模型在其中两种语言中产生了非常不准确的注释，参与者一致认为，纠正预测结果会比从头开始标注或使用更简单的查找方法更加困难。
6. 第三，该模型经常预测出参与者不熟悉或不符合其偏好的注释惯例的注释标签，而现有的系统没有提供任何方法来调整其标签以适应参与者偏好的惯例。

In this work, we address these three concerns, building on the approach of GLOSSLM. We release an improved version of the GLOSSLM corpus, which adds 91k examples for a total of 341k examples, improves standardization and formatting, and ensures alignment between morphological segmentation and glosses. We train a multilingual model on the dataset for joint segmentation and glossing, optimizing for performance on both tasks, as well as alignment between the two tasks, outperforming various small LLMs and satisfying item 1. We show that per-language perplexity can roughly predict glossing accuracy for any language, addressing item 2 by enabling an automatic glossing system to avoid giving low-quality predictions, or to fall back to a simpler model. Finally, we show that POLYGLOSS can be rapidly adapted to small labeled datasets via low-rank adaptation, satisfying item 3. Unlike prior work that trains monolingual glossing models, we focus on creating a single multilingual model that can be used out-of-the-box on many languages. Our models and dataset are available on HuggingFace¹ and our code is on GitHub.²

在本工作中，我们针对这三个问题，基于 GLOSSLM 的方法进行研究。我们发布了一个改进后的 GLOSSLM 语料库版本，新增了 91,000 个例子，总计 341,000 个例子，改进了标准化和格式化，并确保形态分析与释义之间的对齐。我们在该数据集上训练了一个多语言模型，用于联合分词和释义任务，优化了两个任务的性能以及两者之间的对齐，优于各种小型语言模型，并满足了第 1 项要求。我们展示了每种语言的困惑度可以大致预测该语言的释义准确性，通过使自动释义系统避免给出低质量的预测或回退到更简单的模型，从而解决了第 2 项要求。最后，我们展示了 POLYGLOSS 可以通过低秩适应快速适应于小型标注数据集，满足了第 3 项要求。与之前训练单语言释义模型的工作不同，我们专注于创建一个可以开箱即用的单个多语言模型，适用于多种语言。我们的模型和数据集可在 HuggingFace¹ 上获取，我们的代码可在 GitHub

上获取。²

2 Corpus

2 语料库

We create an enhanced version of the GLOSSLM corpus with significantly improved formatting. We consistently handle punctuation across all sources, ensuring that sentence-ending punctuation is surrounded by spaces while gloss-internal punctuation

我们创建了一个增强版的 GLOSSLM 语料库，其格式有显著改进。我们一致处理所有来源中的标点符号，确保句末标点符号前后有空格，而 gloss 内部的标点符号则被正确处理。

Statistic	Count
Total examples	353,266
Unique languages	2,077
Train examples	340,251
Eval examples	6,148
Test examples	6,867
No glottocode	13,428
No metalang. glottocode	10,894
No segmentation	93,648
No translation	5,921
Misaligned	34,894

统计	计数
总示例数	353,266
独特语言	2,077
训练示例	340,251
评估示例	6,148
测试示例	6,867
无音位代码	13,428
无金属语言。glottocode	10,894
不分段	93,648
没有翻译	5,921
错位	34,894

Table 1: POLYGLOSS corpus statistics

表 1: POLYGLOSS 语料库统计信息

remains unchanged, as in the following Tsez example:

保持不变，例如以下的 Tsez 示例：

(1) Žeda kidbeqor kurno lel yayrno . žeda-a kid-qor kur-n lel y-ayr-n DEM1.IIPL.OBL-ERG girl-POSS.LAT throw- PFV.CVB wing II-lead-PST.UNW

(1) 女孩扔掉了她的翅膀。
 žeda-a kid-qor kur-n lel y-ayr-n
 DEM1.IIPL.OBL-ERG girl-POSS.LAT throw-
 PFV.CVB wing II-lead-PST.UNW

We fixed a number of source-specific formatting issues. For example, we noticed 4,882 instances in the Arapaho data where “,” was used inside glosses, and we confirmed with the original annotator that this was an error. Across sources, we identified instances where the morphological segmentation was misaligned with the glosses—that is, cases where there was a mismatch in either the number of words or the number of segments within a word (see §3.3 for more discussion of alignment). In these cases, if the segmentation field does not include any segmentation markers (and the gloss field does), we set the segmentation to blank. Otherwise, we keep the segmentation, but ensure the offending examples are within the training split, so as not to affect evaluation.

我们修复了多个与数据源相关的格式问题。例如，我们在阿拉帕霍语数据中发现了 4,882 个实例，其中在注释中使用了逗号 “,”，我们与原始标注者确认，这确实是一个错误。在跨数据源的分析中，我们识别了形态分割与注释不匹配的实例——也就是说，单词数量或单词内部段落数量不一致的情况（有关对齐的更多讨论，请参见第 3.3 节）。在这些情况下，如果分割字段不包含任何分割标记（而注释字段包含），我们将分割设置为空。否则，我们保留分割，但确保有问题的示例位于训练集内，以免影响评估。

We also incorporate additional IGT data into the original GLOSSLM corpus. The largest of these is the Fieldwork dataset (He et al., 2024), which collects 80,461 IGT instances for 37 languages. We also update the IMTVault dataset (Nordhoff and Krämer, 2022) to the newest version (1.2), which includes IGT scraped from new linguistic publications, adding 39,741 examples. After removing 20,116 duplicates and filtering very low-quality examples,³ we have 91,416 new unique examples compared to the original GLOSSLM corpus. We introduced an auditing process on our dataset to quantify issues, and report full statistics in Table 1. We add two new languages as evaluation languages from the Fieldwork dataset: Hokkaido Ainu

我们还将额外的 IGT 数据纳入原始的 GLOSSLM 数据集。其中最大的一个数据集是实地调查数据集（He 等, 2024），该数据集收集了 37 种语言的 80,461 个 IGT 实例。我们还更新了 IMTVault 数据集（Nordhoff 和 Krämer, 2022）至最新版本（1.2），该版本包括从新的语言学出版物中抓取的 IGT 数据，新增了 39,741 个实例。在删除了 20,116 个重复项并过滤掉质量非常低的实例后，³ 与原始 GLOSSLM 数据集相比，我们新增了 91,416 个独特的实例。我们对数据集进行了审计流程以量化问题，并在表 1 中报告完整的统计数据。我们从实地调查数据集中新增了两种语言作为评估语言：北海道阿伊努语

(ainu1240) and Ruuli (ruul1235). Our dataset splits for all evaluation languages are reported in Table 2.

(ainu1240) 和 Ruuli (ruul1235)。我们所有评估语言的数据集划分结果在表 2 中报告。

Language	Train	Eval	Test
Arapaho (arp)	36776	4687	4499
Tsez (ddo)	3626	444	442
Gitksan (git)	89	42	37
Uspanteko (usp)	8338	170	566
Ainu (ain)	6726	218	590
Lezgi (lez)	646	51	53
Natugu (ntu)	786	99	99
Nyangbo (nyb)	1221	225	248

Ruuli (ruc)	2158	212	333
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语言	火车	评估	测试
阿拉帕霍语 (arp)	36776	4687	4499
泰泽语 (ddo)	3626	444	442
Gitksan (git)	89	42	37
乌斯潘特克语 (Us)	8338	170	566
阿伊努人 (ain)	6726	218	590
勒兹吉语 (lez)	646	51	53
纳图古 (ntu)	786	99	99
尼洋博 (nyb)	1221	225	248
Ruuli (ruc)	2158	212	333

Table 2: Number of examples for each evaluation language across train, eval, and test splits.

表 2: 每个评估语言在训练集、评估集和测试集中的示例数量。

3 Evaluation

3 评估

Since our model performs joint glossing and segmentation, we compute metrics for both, as well as an alignment score (§3.3) between the two.

由于我们的模型同时进行分词和词性标注，我们计算这两个任务的指标，以及两者之间的对齐得分 (§3.3)。

3.1 Glossing

3.1 词性标注

We compute a number of metrics for gloss prediction. Departing from prior work, we use morpheme error rate (c.f. word error rate as used in speech recognition) as our primary metric. While prior work (McMillan-Major, 2020a; Ginn et al., 2023) used morpheme-level accuracy as the primary metric, this assumes that the output has the correct number of morphological glosses. If there is a gloss inserted or deleted, all subsequent glosses will be counted as incorrect. Instead, we compute the morpheme error rate by first inserting [SEP] tokens between the glosses for each word and computing the edit distance, normalized to the length of the gold label sequence. The range is 0 or greater; a score higher than 1 is possible if the predicted sequence is longer than the gold sequence. In addition, we compute word and character error rates, BLEU scores (at all three levels of granularity), and morpheme and word-level accuracy.

我们计算了一系列用于词形标注预测的指标。与之前的工作不同，我们使用形态素错误率（参考语音识别中使用的词错误率）作为我们的主要指标。虽然之前的工作 (McMillan-Major, 2020a; Ginn et al., 2023) 将形态素级别的准确率作为主要指标，但这一假设前提是输出具有正确数量的形态素标注。如果插入或删除了某个标注，那么后续的所有标注都会被计为错误。相反，我们首先在每个词的标注之间插入 [SEP] 标记，然后计算编辑距离，并将其归一化到黄金标签序列的长度上。该指标的取值范围是 0 或更大；如果预

测序列长度超过黄金序列，得分可能会大于 1。此外，我们还计算了词错误率和字符错误率、BLEU 分数（在所有三个粒度级别上），以及形态素和词级别的准确率。

3.2 Segmentation

3.2 分割

We use standard metrics for evaluating segmentation. We primarily report the modified F1 score as defined in Mager et al. (2020), which computes precision based on morphemes in the predicted segmentation also occurring in the gold label, and vice versa for recall. We also compute character-level edit distance and whole-word accuracy.

我们使用标准指标来评估分割效果。我们主要报告由 Mager 等人 (2020) 定义的改进型 F1 分数，该指标基于预测分割中的语素与黄金标签中的语素是否一致来计算精确率，反之亦然用于召回率。我们还计算字符级别的编辑距离和整个单词的准确率。

3.3 Alignment

3.3 对齐

A key goal in this study is to predict morphological segmentations and glosses that are aligned with one another, making the gloss predictions more interpretable and trustworthy for a human annotator. To measure this, we propose a novel alignment score, which is computed based on predictions with no reference to the gold sequence.⁴ First, the segmentation and gloss predictions are converted into abstract sequences that represent morphological structure. Ignoring punctuation, each morpheme sequence is converted to a single “x” character, and morpheme boundaries (“-” and “=”) are left unchanged, as in the following example:

本研究的一个关键目标是预测与彼此对齐的形态学分段和词素标注，从而使词素标注的预测对于人类标注者来说更加可解释和可信。为了衡量这一点，我们提出了一种新颖的对齐得分，该得分是基于预测结果计算得出的，而无需参考黄金序列。⁴ 首先，分段和词素标注的预测结果被转换为表示形态结构的抽象序列。忽略标点符号，每个词素序列被转换为一个“x”字符，而词素边界（“-”和“=”）则保持不变，如下例所示：

the cat - s ru - n $\Rightarrow$ xx — xx — x

DET cat - PL run.SG $\Rightarrow$ xx — xx

Then, the character-level edit distance is computed between the abstracted gloss and segmentation sequences, ranging from 0 to infinity. We normalize the edit distance by the length of the longer sequence,⁵ and subtract from 1 to give a score in $[0, 1]$ where 1 is a perfect score. In this example, the alignment score is 0.78.

然后，计算抽象后的词素和分词序列之间的字符级编辑距离，范围从 0 到无穷大。我们通过将编辑距离除以较长序列的长度⁵，并从 1 中减去以得到一个得分 $[0, 1]$ ，其中 1 表示完美得分。在此示例中，对齐得分是 0.78。

4 Model

4 模型

4.1 Task Format

4.1 任务格式

Using the POLYGLOSS corpus, we perform continued pretraining on a pretrained multilingual LLM for both segmentation and glossing. We train the model to predict glosses from both the segmented and unsegmented transcription, but we only evaluate on the latter, as it is the more difficult and realistic setting. We experiment with three different approaches for combining the two tasks and report results in 7.2.

使用 POLYGLOSS 语料库，我们对一个预训练的多语言大语言模型进行持续预训练，以实现分词和注释任务。我们训练模型从分词和未分词的转录中预测注释，但我们只在后者上进行评估，因为这是更具挑战性和现实的设置。我们尝试了三种不同的方法来结合这两个任务，并在 7.2 节中报告了结果。

Multitask Prediction In this setting, separate training examples are created for segmentation and for glossing. The examples are formatted as in the following Vera’a language example (replacing “glosses” with “segmentation” when appropriate):

多任务预测在这种设置下，为分词和加注分别创建独立的训练示例。这些示例的格式如下所示的 Vera 语言示例（在适当的情况下将“加注”替换为“分词”）：

Predict the glosses for the following text in Vera’a.

预测以下文本在 Vera’a 中的词缀。

Text in Vera’a: o wōlēn ’ēqēk

文纳的话：哦，我饿了

Translation in English: Oh, over there is my garden

哦，那边是我的花园

Glosses: INTERJ you.know-ZERO=ART garden-1SG

注释：INTERJ you.know-ZERO=ART garden-1SG

This setting is simple and allows for simultaneous inference of both glosses and segmentation. How-

此设置简单，并允许同时推断出两个词义和分割。如何-

ever, there is greater risk of misalignment, since the two tasks are trained separately and alignment is not enforced.

然而，存在更大的对齐风险，因为这两个任务是分别进行训练的，且并未强制执行对齐。

Concatenated Prediction Morphological segmentation and interlinear glosses are not in fact distinct tasks, as the latter depends inherently on the former. In this setting, we model this dependency by training the model to predict the segmentation followed by the glosses:

拼接预测形态学分词和逐行注释并不是本质上不同的任务，因为后者本质上依赖于前者。在这种设置下，我们通过训练模型首先预测分词，然后预测注释来建模这种依赖关系：


```
Predict the morphological segmentation and glosses for the following text in Vera' a.
Text in Vera' a: o wōlēn 'ēqēk
Translation in English: Oh, over there is my garden
Segmentation: o wōlē-Ø=n 'ēqē-k
Glosses: INTERJ you.know-ZERO=ART garden-1SG
```

This introduces a natural dependency due to the causal training objective: while generating the gloss string, the model can attend to tokens in the segmentation. Of course, this is not a strict constraint, and carries the risk that a bad segmentation will affect the glosses as well. Since not all training examples have segmentation labels, we also create glossing examples in the multitask style shown in the previous section.

这引入了一个自然的依赖关系，这是由于因果训练目标所导致的：在生成词形变化字符串时，模型可以关注分词中的标记。当然，这并不是一个严格的限制，而且存在这样的风险：一个不好的分词会影响词形变化。由于并非所有的训练示例都有分词标签，我们还按照前一节中所示的多任务风格创建了词形变化示例。

Interleaved Prediction While the concatenated setting trains the model with an implicit relationship between segments and glosses, it is still possible to generate misaligned predictions. In this setting, we introduce a hard constraint using an interleaved format that explicitly aligns segments and glosses. In this format, each gloss label is immediately followed by the corresponding morpheme in parentheses.

交错预测在串联设置中，模型是通过段落和词素之间的隐含关系进行训练的，但仍有可能生成对齐错误的预测。在这种设置下，我们引入一种硬性约束，使用交错格式明确地对齐段落和词素。在这种格式中，每个词素标签后立即跟随对应的语素，用括号括起来。

```
Predict the glosses and morphological segmentation (in parentheses) for the following te
Text in Vera' a: o wōlēn 'ēqēk
Translation in English: Oh, over there is my garden
Output: INTERJ(o) you.know(wōlē)-ZERO(Ø)=ART(n) garden('ēqē)-1SG(k)
```

We hypothesized that this setting would have the best alignment, as any well-formed output should be perfectly aligned.

我们假设这种设置会有最佳的对齐效果，因为任何格式良好的输出都应该完全对齐。

4.2 Base Model

4.2 基础模型

Following (Ginn et al., 2024b), we perform continued pretraining on ByT5 (Xue et al., 2022), a byte-level encoder-decoder transformer language model based on the T5 architecture. By using byte-level tokenization, ByT5 avoids the issues that arise for rare languages with subword tokenizers, and has been shown to outperform T5 on multilingual glossing (He et al., 2023). We also experimented

按照 (Ginn 等人, 2024b) 的方法，我们在基于 T5 架构的字节级编码器-解码器 Transformer 语言模型 ByT5 (Xue 等人, 2022) 上进行持续预训练。通过使用字节级分词，ByT5 避免了在使用子词分词器时罕见语言所面临的问题，并且已被证明在多语言标注任务中优于 T5 (He 等人, 2023)。我们还进行了实验

with finetuning an instruction-tuned decoder LLM, using Qwen 3 0.6B (Yang et al., 2025a), known to be a strong multilingual model. However, we saw poor results, discussed in Appendix A. We train one ByT5-based models for each task format in the preceding section, using the byt5-base checkpoint with 580M parameters.

通过微调一个指令调优的解码器大语言模型，使用 Qwen 3 0.6B (Yang 等, 2025a)，该模型被认为是一

个强大的多语言模型。然而，我们得到了较差的结果，详见附录 A。我们在上一节中针对每种任务格式训练了一个基于 ByT5 的模型，使用了具有 580M 参数的 byt5-base 检查点。

4.3 Training

4.3 训练

Due to the high cost of training runs, we do not perform extensive tuning. We train all models in bf16, using the AdamW optimizer with default parameters, a linear learning rate warmup for the first 3% of steps, cosine learning rate decay, and gradient clipping with max norm of 1. We train all models using 4 GH200 GPUs. For the ByT5-based models, we used a learning rate of 5E-5, batch size of 64, and 15 epochs. Evaluation uses beam search with default parameters and 2 beams. Parameters for the Qwen-based model are given in Appendix A.

由于训练运行的成本较高，我们并未进行广泛的调优。我们使用 bf16 训练所有模型，采用 AdamW 优化器并使用默认参数，前 3% 的步骤使用线性学习率预热，学习率衰减采用余弦衰减，梯度裁剪使用最大范数为 1。我们使用 4 个 GH200 GPU 训练所有模型。对于基于 ByT5 的模型，我们使用了学习率 5E-5、批量大小 64 和 15 个 epoch。评估使用默认参数的束搜索，且使用 2 束。基于 Qwen 的模型的参数在附录 A 中给出。

5 Baselines

5 基线

5.1 Multilingual Baselines

5.1 多语言基线

We compare our best model to the following multilingual models for glossing:

我们将我们的最佳模型与以下用于标注的多语言模型进行比较：

GLOSSSLM We compare glossing performance to the pretrained GLOSSLM model, primarily to ensure that incorporating segmentation does not cause glossing performance to regress. As this model was not trained for segmentation, we cannot compare across all metrics. Additionally, the original GLOSSLM model did not include all of the segmented training data, so this is not a perfect head-to-head comparison.

GLOSSSLM 我们将 glossing 性能与预训练的 GLOSSLM 模型进行比较，主要目的是确保引入分段不会导致 glossing 性能下降。由于该模型并未为分段任务进行训练，因此我们无法在所有指标上进行比较。此外，原始的 GLOSSLM 模型并未包含所有分段后的训练数据，因此这并不是一个完美的直接对比。

In-Context Learning Following Ginn et al. (2024a), we use LLMs to predict both the segmentation and glosses in a single pass using the interleaved format. We retrieve ten similar examples (via chrF score) from the training set to provide in-context. We test three models: the Qwen 3 0.6B model with thinking (Yang et al., 2025a) and the Cohere Aya Expanse 8B model (Dang et al., 2024) and Google Gemma 3 4B model (Team, 2025) without thinking. These enable a direct comparison between our finetuned seq2seq models and ICL with LLMs of similar size.

按照 Ginn 等人 (2024a) 的方法，我们使用大语言模型 (LLMs) 以交错格式在单次推理中预测分词和词性标注。我们从训练集中检索十个相似的示例 (通过 chrF 分数进行评估)，以提供上下文信息。我们测试了三个模型：带有思维的 Qwen 3 0.6B 模型 (Yang 等人, 2025a)，Cohere Aya Expanse 8B 模型 (Dang 等人, 2024) 以及不带思维的 Google Gemma 3 4B 模型 (Team, 2025)。这些测试使我们能够直接比较我们的微调序列到序列模型与使用相似规模的 LLMs 进行的上下文学习 (ICL) 之间的性能。

5.2 Monolingual Baselines

5.2 单语基线

In addition, we run preliminary experiments using monolingual models to understand the effect of different approaches to joint training and monolingual versus multilingual training. We use the following baselines, and provide comparisons in 7.1 and 7.2:

此外，我们还使用单语模型运行初步实验，以了解不同联合训练方法以及单语与多语训练效果的影响。我们采用以下基线，并在 7.1 和 7.2 中提供比较：

Finetuned ByT5 (separate) For each language in our test set, we train monolingual models on ByT5 for segmentation and glossing in the Multi-task format, using separate models for each task. We use the same hyperparameters as our pretrained model, except we set a max 30 epochs and use early stopping with patience 5 to prevent overfitting. We do not perform extensive tuning.

通过 ByT5 微调（独立）对于我们测试集中的每种语言，我们使用 ByT5 在多任务格式下分别训练单语模型，用于分词和加注。每个任务使用独立的模型。我们使用与预训练模型相同的超参数，除了将最大训练轮数设为 30 轮，并使用耐心值为 5 的早停机制以防止过拟合。我们不进行广泛的调优。

Finetuned ByT5 (joint) In addition, we train monolingual models for joint segmentation and glossing using the Interleaved format and the same hyperparameters as the prior baseline.

通过 ByT5 进行微调（联合）此外，我们还使用交错格式和与先前基线相同的超参数，训练用于联合分词和加注的单语模型。

Pipeline We train monolingual pipelines of two ByT5 models, where the first model predicts the morphological segmentation and the second model predicts glosses given segments. In this baseline, there is clear risk of error propagation, as incorrect segmentation likely results in incorrect glosses.

流水线我们训练了两个 ByT5 模型的单语流水线，其中第一个模型预测形态学分割，第二个模型根据分割结果预测术语。在此基线中，存在明显的错误传播风险，因为不正确的分割很可能会导致不正确的术语。

Hard Attention Transformer Following Girrbach (2023a), we train monolingual hard attention transformers using straight-through gradient estimation. The model is only trained on glosses, but generates an explicit latent segmentation based on the hard attention between glosses and characters in the input.

Hard Attention Transformer 依照 Girrbach (2023a)，我们使用直通梯度估计来训练单语硬注意力变压器。该模型仅在词素上进行训练，但会根据输入中词素与字符之间的硬注意力生成一个显式的潜在分段。

6 Results

6 结果

We compare our best POLYGLOSS model (ByT5 architecture and Interleaved format) with multilingual baselines on the test set for glossing (Table 3), segmentation (Table 4), and alignment (Table 5). Results for all other metrics, such as BLEU score, morpheme accuracy, and word-level scores are available on our GitHub.

我们在测试集上将我们的最佳 POLYGLOSS 模型（ByT5 架构和交错格式）与多语言基线模型进行比较，用于词形标注（表 3）、分词（表 4）和对齐（表 5）。其他所有指标的结果，如 BLEU 分数、语素准确率和词级评分，可在我们的 GitHub 上找到。

Overall, the multilingual POLYGLOSS model is state-of-the-art on glossing and very strong on segmentation. It outperforms GLOSSLM for glossing on the three overlapping evaluation languages,

likely due to both better dataset preprocessing and improved training hyperparameters. It also far outperforms in-context learning approaches with LLMs that are orders of magnitude larger (220M vs 0.6B-8B). The smallest LLM (Qwen 0.6B) often struggles to conform to the desired format, resulting in misaligned outputs. The larger models (Gemma 4B and Aya 8B) have much higher alignment scores, but still struggle to perform glossing or segmentation accurately. More details on specific types of failures are described in Appendix C. While significantly larger LLMs might show better results, our model is clearly state-of-the-art given constraints on both model size and training budget.

总体而言，多语言的 POLYGLOSS 模型在标注任务上处于领先水平，并且在分词任务上表现非常出色。它在三种重叠评估语言上的标注任务上优于 GLOSSLM，这可能是由于更好的数据集预处理和改进的训练超参数所致。此外，它在上下文学习方法上也远远优于使用规模大几个数量级的 LLM (220M 对比 0.6B-8B)。最小的 LLM (Qwen 0.6B) 经常难以符合所需的格式，导致输出对齐错误。更大的模型 (Gemma 4B 和 Aya 8B) 具有更高的对齐分数，但仍难以准确地进行标注或分词。关于特定类型失败的更多细节在附录 C 中描述。虽然显著更大的 LLM 可能会表现出更好的结果，但考虑到模型规模和训练预算的限制，我们的模型显然是最先进的。

6.1 Predicting Performance on Other Languages

6.1 预测其他语言上的性能

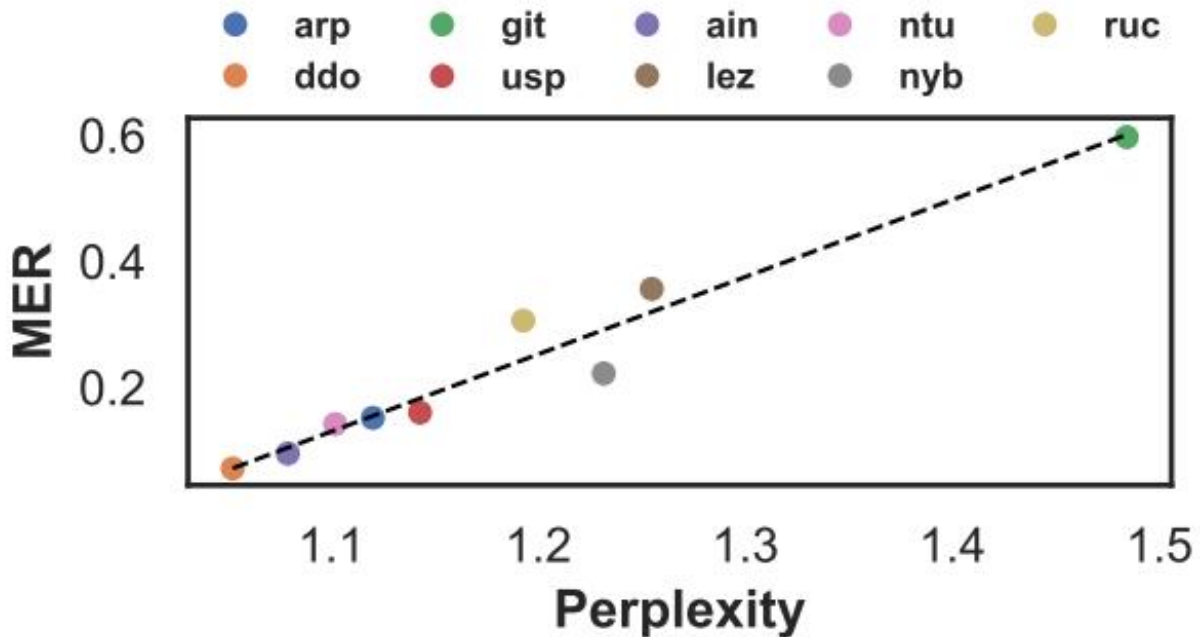


Figure 2: images/91ca092856288c4cff0b901ebfd292c26f67c4def8c6934d381a8ef3098ae609.jpg

Figure 2: Relationship between validation set perplexity for a given language and glossing performance, as measured by morpheme error rate. There is a strong correlation ($r^2 = 0.951$), indicating that perplexity can be used as a rough predictor of glossing performance.

图 2：给定语言的验证集困惑度与释义性能之间的关系，以音位错误率衡量。存在很强的相关性 ($r^2 = 0.951$)，这表明困惑度可以作为释义性能的粗略预测指标。

As identified in Rice et al. (2025), an issue with GLOSSLM was that there was no good way to predict glossing performance on a given language that is not one of our nine selected evaluation languages. For the POLYGLOSS model (ByT5, Interleaved), we compute per-language perplexity

on our validation dataset and demonstrate that it has a strong correlation $r^2 = 0.951$ with our target metric (Figure 2).

如 Rice 等人 (2025) 所指出的, GLOSSLM 的一个问题是, 对于不是我们九种选定评估语言之一的语言, 没有很好的方法来预测其标注性能。对于 POLYGLOSS 模型 (ByT5, Interleaved), 我们在验证数据集上计算每种语言的困惑度, 并展示了它与我们的目标指标有很强的相关性 $r^2 = 0.951$ (图 2)。

This provides a practical heuristic for the use of our model in real-world settings. A glossing software such as Plaid ⁶ could set an acceptable error rate threshold, and use the POLYGLOSS model to predict glosses only if the language’s expected error rate is below that threshold. If not, then it will likely be a better user experience to either fall back to a simple method (such as predicting the highest-frequency gloss) or not showing predictions at all.

这为我们在实际环境中使用该模型提供了一个实用的启发式方法。例如, 像 Plaid ⁶ 这样的标注软件可以设定一个可接受的错误率阈值, 并在语言的预期错误率低于该阈值时, 使用 POLYGLOSS 模型来预测标注。如果不符合, 则可能更好的用户体验是回退到一个简单的方法 (例如预测最高频率的标注) 或者完全不显示预测结果。

	arp	ddo	git	usp	ain	lez	ntu	nyb	ruc	Avg.
Qwen 3 0.6B (ICL)	0.868	0.904	0.919	0.730	0.773	0.895	0.877	0.706	0.883	0.839
Gemma 3 4B (ICL)	0.489	0.597	0.826	0.476	0.351	0.668	0.473	0.430	0.723	0.559
Aya Ex- panse 8B (ICL)	0.545	0.749	0.871	0.514	0.464	0.740	0.591	0.492	0.802	0.641
GLOSSLM161	0.095	0.095	0.870*	0.163	0.909*	0.940*	0.893*	0.990*	0.731*	0.639*
POLYGLOSS177 (ByT5, multi- task)	0.177	0.089	0.603	0.162	0.122	0.383	0.189	0.328	0.329	0.265
POLYGLOSS152 (ByT5, inter- leaved)	0.152	0.072	0.597	0.160	0.095	0.357	0.142	0.222	0.306	0.234

ARP (地址 解析协 议) 是 一种用 于将网 络设备 的 IP 地址转 换为物 理地址 (MAC 地址) 的协 议。它 在局域 网中使 用, 以 使设备 能够通 过 IP 地址找 到彼 此。当 设备需 要与另 一个设 备通信 时, 它 会使用 ARP 来 查找目 标设备 的 MAC 地址。 这个过 程涉及 发送一 个 ARP 请求, 该请求 广播到 网络上的所有 设备, 询问目 标 IP 地址的 MAC 地址。 收到请 求的设 备会检 查是否 与目标 IP 地 址匹 配, 如 果是, 则发送 ARP 响 应, 包 含其	ddo	git	USP	ain	lez	ntu	nyb	ruc	平均值
--	-----	-----	-----	-----	-----	-----	-----	-----	-----

Qwen 3 0.6B (ICL)	0.868	0.904	0.919	0.730	0.773	0.895	0.877	0.706	0.883	0.839
Gemma 3 4B (ICL)	0.489	0.597	0.826	0.476	0.351	0.668	0.473	0.430	0.723	0.559
Aya Ex- panse 8B (ICL)	0.545	0.749	0.871	0.514	0.464	0.740	0.591	0.492	0.802	0.641
GLOSSLM	0.161	0.095	0.870*	0.163	0.909*	0.940*	0.893*	0.990*	0.731*	0.639*
多语言 (ByT5, 多任务)	0.177	0.089	0.603	0.162	0.122	0.383	0.189	0.328	0.329	0.265
多语言 (ByT5, 交错)	0.152	0.072	0.597	0.160	0.095	0.357	0.142	0.222	0.306	0.234

Table 3: Morpheme error rate ($\downarrow$) for glossing on the held-out test set with multilingual models. For GLOSSLM, the only eval languages explicitly included in the pretraining corpus are arp, ddo, and git, so scores on other languages (marked with *) are very poor.

表 3: 在多语言模型上对保留测试集进行标注时的语素错误率 ($\downarrow$)。对于 GLOSSLM, 唯一在预训练语料库中明确包含的语言是 arp、ddo 和 git, 因此其他语言 (标记为 * 的语言) 的得分非常差。

	arp	ddo	git	usp	ain	lez	ntu	nyb	ruc	Avg.
Qwen 3 0.6B (ICL)	0.091	0.096	0.050	0.297	0.233	0.118	0.133	0.298	0.190	0.167
Gemma 3 4B (ICL)	0.505	0.310	0.159	0.508	0.575	0.338	0.435	0.617	0.345	0.421
Aya Ex- panse 8B (ICL)	0.457	0.249	0.132	0.509	0.569	0.302	0.337	0.465	0.324	0.371
POLYGLOSS (ByT5, multi- task)	0.903	0.967	0.605	0.850	0.963	0.826	0.925	0.938	0.763	0.860
POLYGLOSS (ByT5, inter- leaved)	0.910	0.972	0.595	0.855	0.963	0.782	0.935	0.959	0.782	0.862

ARP (Ad- dress Reso- lution Proto- col, 地 址解析 协议) 是一种 用于将 网络层 地址 (如 IP 地址) 转换为 数据链 路层地 址 (如 MAC 地址) 的协 议。它 在局域 网 (LAN) 中广泛 使用, 以确保 数据能 够正确 地从一个设备 传输到 另一个 设备。 ### ARP 的 工作原 理 1. 请求: 当一个 设备需 要发送 数据给 另一个 设备 时, 它 会先检 查自己 的 ARP 缓存, 看看是 否已经 有目标 设备的 MAC 地址。 2. 广 播: 如 果 ARP 缓 存中没 有目标	DDO	git	USP	ain	lez	ntu	nyb	ruc	平均值
---	-----	-----	-----	-----	-----	-----	-----	-----	-----

Qwen 3 0.6B (ICL)	0.091	0.096	0.050	0.297	0.233	0.118	0.133	0.298	0.190	0.167
Gemma 3 4B (ICL)	0.505	0.310	0.159	0.508	0.575	0.338	0.435	0.617	0.345	0.421
Aya Ex- panse 8B (ICL)	0.457	0.249	0.132	0.509	0.569	0.302	0.337	0.465	0.324	0.371
多语言 (ByT5, 多任务)	0.903	0.967	0.605	0.850	0.963	0.826	0.925	0.938	0.763	0.860
多语言 (ByT5, 交错)	0.910	0.972	0.595	0.855	0.963	0.782	0.935	0.959	0.782	0.862

Table 4: Morpheme F1 (↑) for segmentation on the held-out test set with multilingual models.

表 4: 多语言模型在保留测试集上的分词 F1 (↑) 指标。

	arp	ddo	git	usp	ain	lez	ntu	nyb	ruc	Avg.
Qwen 0.6B (ICL)	0.342	0.669	0.488	0.744	0.716	0.711	0.697	0.788	0.792	0.661
Gemma 3 4B (ICL)	0.982	0.995	0.936	0.983	0.985	0.993	0.994	0.993	0.990	0.984
Aya Ex- panse 8B (ICL)	0.957	0.953	0.956	0.973	0.979	0.882	0.978	0.985	0.986	0.961
POLYGLO (ByT5, multi- task)	0.984	0.995	0.919	0.990	0.982	0.973	0.985	0.988	0.941	0.973
POLYGLO (ByT5, inter- leaved)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

ARP (地址 解析协 议) 是 一种用 于将网 络设备 的 IP 地址转 换为物 理地址 (MAC 地址) 的协 议。它 在局域 网中使 用, 以 确定数 据包应 发送到 哪个物 理设 备。当 设备需 要与另 一个设 备通信 时, 它 会使用 ARP 来 查找目 标设备 的 MAC 地址。 这通常 通过发 送一个 ARP 请 求广播 到网络 中的所有 设备, 然 后目标 设备会 回复其 MAC 地址。 一旦获 得 MAC 地址, 源设备 就可以 直接将 数据包 发送到 目标设 备。	DDO	git	USP	ain	lez	ntu	nyb	ruc	平均值
---	-----	-----	-----	-----	-----	-----	-----	-----	-----

Qwen 0.6B (ICL)	0.342	0.669	0.488	0.744	0.716	0.711	0.697	0.788	0.792	0.661
Gemma 3 4B (ICL)	0.982	0.995	0.936	0.983	0.985	0.993	0.994	0.993	0.990	0.984
Aya Ex- panse 8B (ICL)	0.957	0.953	0.956	0.973	0.979	0.882	0.978	0.985	0.986	0.961
多语言 (ByT5, 多任务)	0.984	0.995	0.919	0.990	0.982	0.973	0.985	0.988	0.941	0.973
多语言 (ByT5, 交错)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 5: Alignment score ($\uparrow$) between predicted segmentation and glosses on held-out test set, multilingual models.

表 5：在保留测试集上的对齐得分（ $\uparrow$ ），预测分割与词素之间的对齐得分，多语言模型。

7 Ablations

7 消融实验

Per-language breakdowns are provided for all ablations in Appendix B.

按语言划分的分析结果在附录 B 中提供。

7.1 Effect of Joint Training

7.1 联合训练的效果

The best POLYGLOSS model is trained jointly on segmentation and glosses in a single training example. To disentangle the effect of this joint training approach, we compare three monolingual ByT5-based approaches: training for glossing and segmentation separately, jointly, and in a model pipeline (see subsection 5.2). We also evaluate the Girrbach (2023a) approach, which induces a latent segmentation using hard attention. We report the average scores in Figure 3. ⁷ The Joint setting is superior on all three metrics (and near-perfect on alignment), suggesting a harmonious relationship between the two training tasks. While the Separate setting is similar to the former on glossing and

最好的 POLYGLOSS 模型是在一个训练示例中联合训练分词和词性标注的。为了分离这种联合训练方法的影响，我们比较了三种基于单语 ByT5 的方法：分别进行词性标注和分词训练、联合训练以及在模型流水线中训练（参见子节 5.2）。我们还评估了 Girrbach (2023a) 的方法，该方法使用硬注意力诱导出潜在的分词。我们在图 3 中报告了平均得分。⁷ 在所有三个指标（以及接近完美的对齐度）上，联合设置的表现都优于其他设置，这表明这两个训练任务之间存在和谐的关系。虽然单独设置在词性标注和

segmentation, its alignment score is significantly worse. This is unsurprising, as the two separate

models for glossing and alignment are not guaranteed to produce the same errors, and thus may generate misaligned outputs for the same input. For example, the separated models make the following poorly aligned prediction for a Nyangbo sentence (alignment score of 0.882):

分词，其对齐得分显著更差。这是意料之中的，因为用于注释和对齐的两个独立模型并不能保证产生相同的错误，因此可能会对同一输入生成不一致的输出。例如，分离的模型对一个 Nyangbo 句子做出以下对齐效果不佳的预测（对齐得分为 0.882）：

(2) vūnə gagālī gɛ enu budzyuɖí yɛ vūnə gagālī gɛ e-nu bu-dzyuɖí yɛ beverage well-well REL 3SG-be CM-strength 3SG

(2) vūnə gagālī gɛ enu budzyuɖí yɛ
vūnə gagālī gɛ e-nu bu-dzyuɖí yɛ
beverage well-well REL 3SG-be CM-strength 3SG

The misalignment occurs in the second word, which is predicted to be a single morpheme but two glosses. Meanwhile, the interleaved joint model predicts the perfectly aligned glosses and morphemes (which are also more accurate):

错位出现在第二个词中，该词被预测为一个语素但有两个注释。与此同时，交错联合模型预测了完全对齐的注释和语素（这也更准确）：

(3) vūnə gagālī gɛ enu budzyuɖí yɛ vūnə gagālī gɛ e-nu bu-dzyuɖí yɛ greet be_hard REL 3SG-be CM-dawadawa_tree FOC

(3) vūnə gagālī gɛ enu budzyuɖí yɛ
vūnə gagālī gɛ e-nu bu-dzyuɖí yɛ
greet be_hard REL 3SG-be CM-dawadawa_tree FOC

For real-world usage, generating aligned outputs is critical in addition to achieving high accuracy.

在实际应用中，除了实现高准确性外，生成对齐的输出也至关重要。

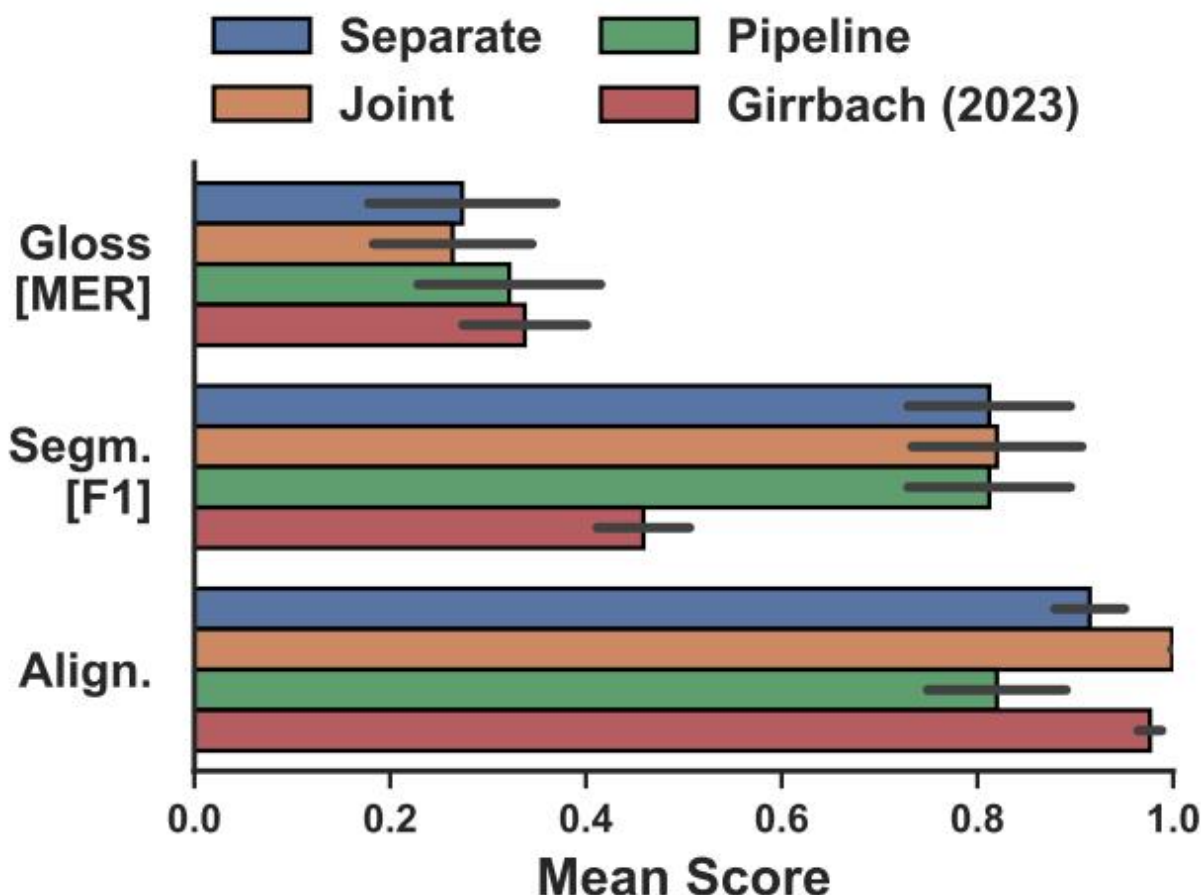


Figure 3: images/03a4e28fe0d76cc5386bd86152fa25256b390a2bf9536d76b916298af7c6b8cc.jpg

Figure 3: Scores for monolingual models using various approaches to multitask training. A lower glossing MER is better; higher is better for the other two metrics. Scores are averaged across nine languages and reported with standard error.

图 3：使用不同多任务训练方法的单语模型得分。较低的词缀标注 MER 值更好；另外两个指标较高的值更好。得分是在九种语言上平均得到的，并带有标准误差。

The Joint approach is also clearly superior to the Pipeline setting for glossing and alignment (though segmentation is very similar). Because the pipeline is not trained end-to-end, error in the intermediate predictions is catastrophic for the gloss predictions, resulting in far higher error than the joint approach. Finally, the jointly trained model outperforms the hard attention approach of Girrbach (2023a) on segmentation, where the learned segmentations are not very accurate to the gold labels.

联合方法在注释和对齐任务中也明显优于管道设置（尽管分段非常相似）。由于管道方法不是端到端训练的，中间预测中的错误会对注释预测造成灾难性影响，导致错误率远高于联合方法。最后，在分段任务中，联合训练的模型在分段效果上优于 Girrbach (2023a) 的硬注意力方法，其中学习到的分段结果与黄金标签并不十分准确。

7.2 Effect of Multilingual Pretraining

7.2 多语言预训练的效果

We compare the monolingual joint models with the POLYGLOSS multilingual model (Figure 4). Both approaches use the interleaved format, enabling a direct comparison. We see that on average, the multilingual model outperforms the monolingual models on glossing and segmentation (and alignment is perfect for both), though the standard error is large. We observed that for the three languages with the most training data (arp, usp, and ain) the monolingual model is slightly superior on glossing. Meanwhile, for all other languages the multilingual model is far superior on glossing, demonstrating clear transfer learning benefits. Furthermore, we believe that serving a single multilingual glossing model is preferable to requiring linguists to train and host their own monolingual models (though we do propose an adaptation method in section 8).

我们将单语联合模型与 POLYGLOSS 多语言模型进行比较（图 4）。两种方法都使用交错格式，从而可以进行直接比较。我们发现，在平均情况下，多语言模型在注释和分段任务中优于单语模型（并且两种方法的对齐效果都完美），尽管标准误差较大。我们观察到，对于训练数据最多的三种语言（arp、usp 和 ain），单语模型在注释任务上略胜一筹。与此同时，对于所有其他语言，多语言模型在注释任务上表现显著更好，这表明了明显的迁移学习优势。此外，我们认为提供一个单一的多语言注释模型比要求语言学家训练和托管自己的单语模型更可取（尽管我们在第 8 节中提出了一种适应方法）。

7.3 Effect of Task Format

7.3 任务格式的影响

Results for the three ByT5-based POLYGLOSS models are very similar regardless of task format

三种基于 ByT5 的 POLYGLOSS 模型在任务格式不同的情况下，结果非常相似

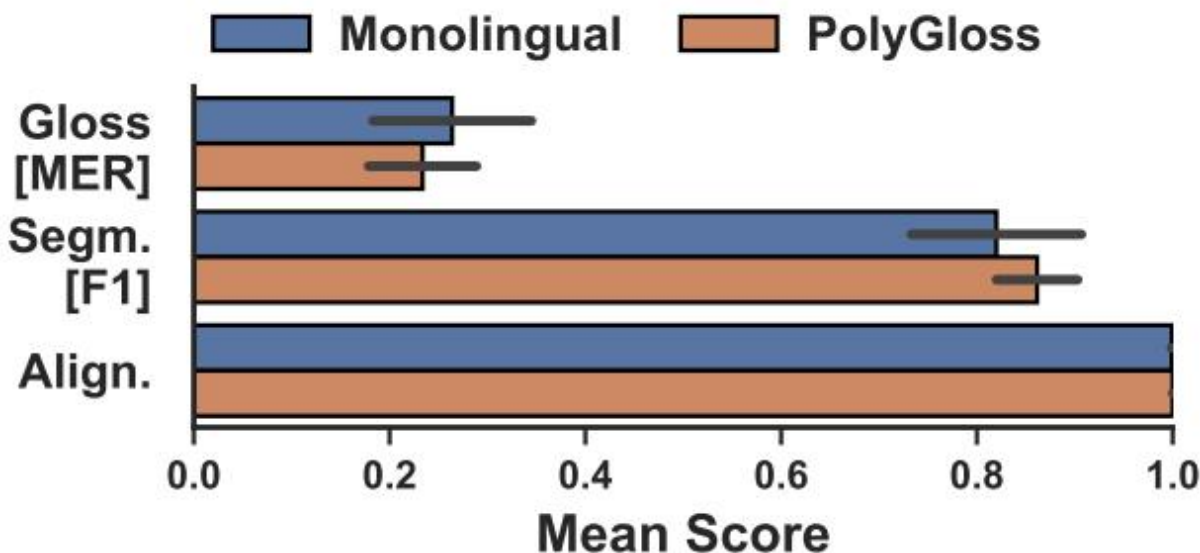


Figure 4: images/4b8e086e52d1c47c2d2028eebc9ad074b1e9c2a00e35a1c6b189de485e3b901b.jpg

Figure 4: Ablation comparing monolingual joint models (using the interleaved format) and the multilingual POLYGLOSS using the same format. A lower glossing MER is better; higher is better for the other two metrics. Scores are averaged across nine languages and reported with standard error.

图 4: 对比使用交错格式的单语联合模型和使用相同格式的多语 POLYGLOSS 的消融实验。较低的 glossing MER 表现更好；对于另外两个指标，较高的值更好。分数是在九种语言上平均计算的，并附有标准误差。

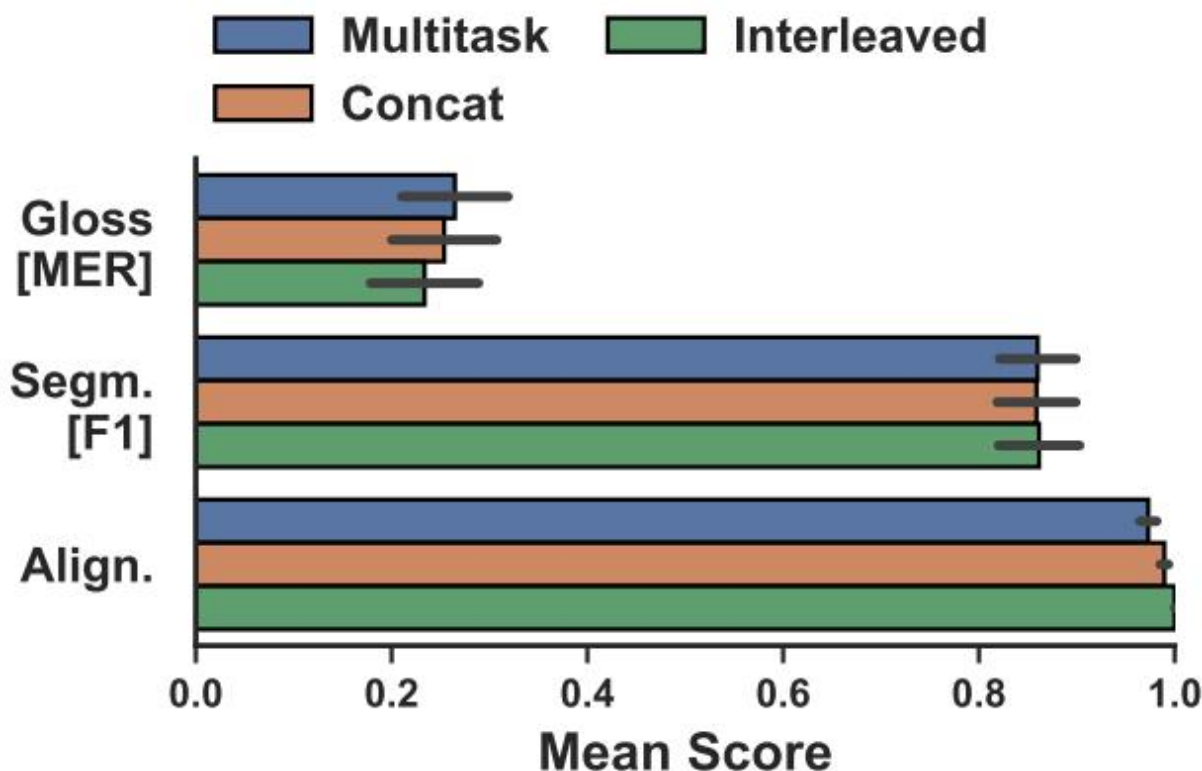


Figure 5: images/d72edf9da106325f822478cd14ad6e246bbd15a04bcd44292f58d57267699d9.jpg

Figure 5: Scores for POLYGLOSS multilingual models using three different data formats. Scores are averaged across nine languages and reported with standard error.

图 5: 使用三种不同数据格式的 POLYGLOSS 多语言模型的得分。得分在九种语言上进行平均，并带有标准误差。

(Figure 5). The interleaved model is best overall on glossing (average MER 0.234), segmentation (average F1 0.862), and alignment (1.000). The multitask model is slightly worse at glossing, indicating potential benefits from explicitly conditioning the model on the morphological segmentation. As hypothesized, the multitask model struggles the most on alignment, indicating that its glossing and segmentation predictions are accurate but not necessarily aligned. The interleaved format results in perfect alignment thanks to its explicit constraint.

(图 5)。在标注（平均 MER 0.234）、分词（平均 F1 0.862）和对齐（1.000）任务上，交错模型表现最好。多任务模型在标注任务上略逊一筹，这表明显式地将模型条件化在形态学分词上可能有潜在优势。正如所预期的那样，多任务模型在对齐任务上表现最差，这表明其标注和分词预测虽然准确，但不一定对齐。由于交错格式具有显式的约束，因此能够实现完美的对齐。

One concern with the interleaved and concatenated formats is that the expected output is roughly twice as long than the joint model, as it includes both the glosses and segmentation. Due to the autoregressive nature of generation, longer sequences can result in degraded accuracy. We test this by filtering the test set to only very long inputs; specifically, we only keep examples where the transcription length is in the 75th percentile for all examples of the same language. We compare glossing re-

一种对交错和连接格式的担忧是，预期的输出长度大约是联合模型的两倍，因为它包括了词素和分词结果。

由于生成的自回归性质，更长的序列可能导致准确性下降。我们通过过滤测试集来测试这一点，只保留非常长的输入；具体来说，我们只保留那些转录长度处于同一语言所有示例的第 75 个百分点的示例。我们比较词素重生成的性能。

sults for long inputs between the three formats in Figure 6. We observe that average performance is nearly identical on long inputs, thus suggesting there is no concern with the increased length of the interleaved and concatenated formats.

结果表明，在图 6 中三种格式之间，对于长输入的性能差异不大。我们观察到，在长输入上，平均性能几乎相同，这表明增加的交错和连接格式的长度不会带来显著的问题。

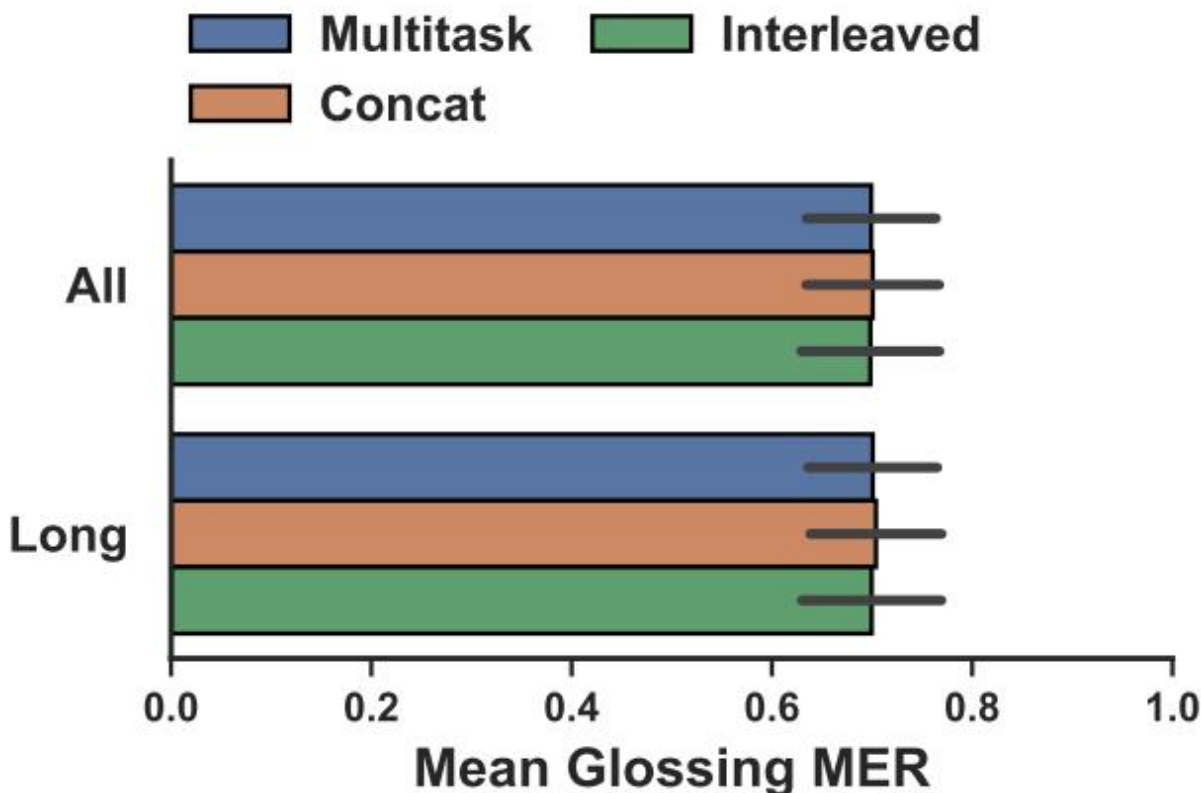


Figure 6: images/4c1d804c2d3826b1a0afae4871d0b0059291e9a1dd29e11882d1b042a545f4ec.jpg

Figure 6: Glossing MER for POLYGLOSS multilingual models when evaluated either on all data, or only on long inputs (75th percentile in length per language). Full data in Table 10.

图 6：在评估 POLYGLOSS 多语言模型时，对所有数据或仅对长输入（每种语言长度的 75 百分位数）进行标注的 MER。完整数据见表 10。

8 Adapting POLYGLOSS To New Data

8 适应新数据的 POLYGLOSS

Though we strived to collect as much IGT data as possible, it is inevitable that some languages will not be well-represented (or present at all) in the POLYGLOSS pretraining corpus. Furthermore, as glossing conventions vary between annotators, the glosses produced by POLYGLOSS may not match the desired schema. Ginn et al. (2024b) studied full-parameter finetuning for unseen languages, but we argue this is an unrealistic scenario for virtually all documentary linguists (due to both technical difficulty and compute requirements). Instead, we propose the use

of low-rank adaptation (LoRA), which drastically reduces the computational cost of training and often requires less data (Hu et al., 2022). Given a small dataset of new glossed examples, it is feasible to run LoRA adaptation with limited computational resources. For example, a glossing software could periodically train an adapter overnight, enabling predictions in the linguist’s target language without any additional effort.

尽管我们努力收集尽可能多的 IGT 数据，但不可避免地，某些语言在 POLYGLOSS 预训练语料库中可能没有得到很好的代表（甚至完全缺失）。此外，由于标注者之间的注释惯例不同，POLYGLOSS 生成的注释可能与期望的模式不匹配。Ginn 等人（2024b）研究了对未见过的语言进行全参数微调，但我们认为对于几乎所有的语言学家来说，这种情况是不现实的（由于技术难度和计算需求）。相反，我们建议使用低秩适应（LoRA），这大幅降低了训练的计算成本，并且通常需要更少的数据（Hu 等人，2022）。给定一小部分新的注释示例数据集，使用有限的计算资源运行 LoRA 适应是可行的。例如，一个注释软件可以定期在夜间训练一个适配器，从而在语言学家的目标语言中进行预测，而无需任何额外的努力。

We simulate a realistic annotation scenario for Vamale (which is never seen during pretraining), training LoRA adapters on increasingly large training datasets (increasing increments of 50 up to the full 380 examples). We use the Vamale data from Yang et al. (2025b) and train rank 8 adapters for 25 epochs, using a batch size of 32; on an A100, the largest training only took 12 minutes. We re-

我们模拟了一个现实的标注场景，用于 Vamale（该数据集在预训练过程中从未见过），在逐渐增大的训练数据集上训练 LoRA 适配器（增量增加 50，直到完整的 380 个示例）。我们使用了 Yang 等人（2025b）提供的 Vamale 数据集，训练了秩为 8 的适配器，进行了 25 个 epoch 的训练，使用了 32 的批次大小；在 A100 显卡上，最大的训练仅耗时 12 分钟。我们重新

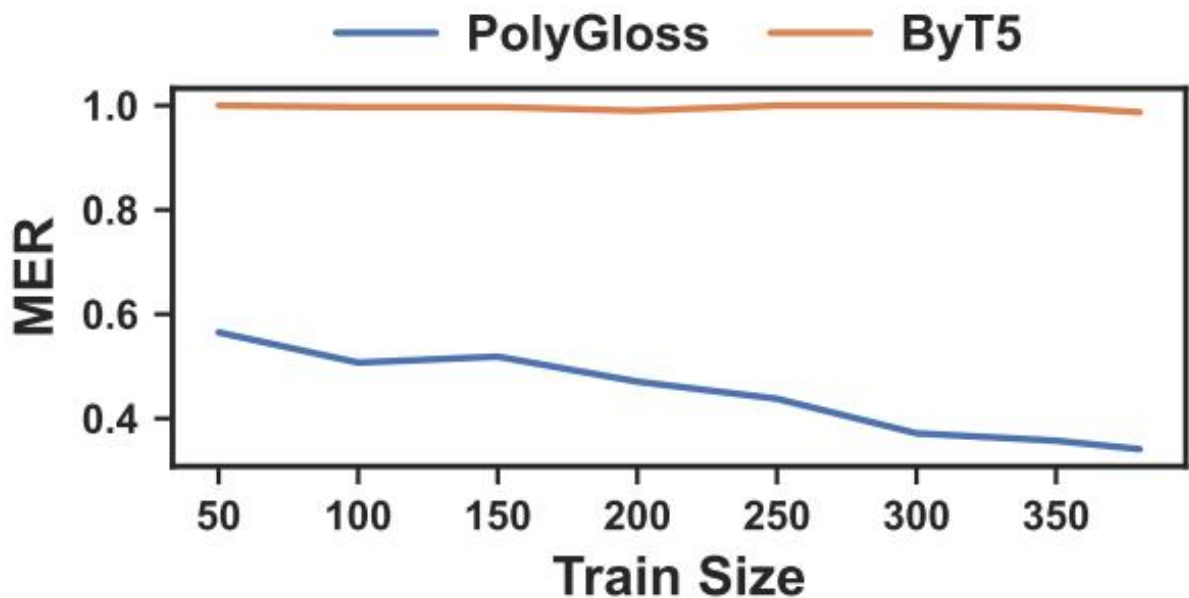


Figure 7: images/094f486b3442d427404256e72cb84f834ed24b820331d592bad6b5778fb996b4.jpg

Figure 7: Morpheme error rate for Vamale when training LoRAs on the POLYGLOSS interleaved model and ByT5 with different size training sets

图 7：在使用不同规模训练集对 POLYGLOSS 交错模型和 ByT5 进行 LoRAs 训练时，Vamale 的语素错误率

port MER scores when adapting the interleaved POLYGLOSS model and a ByT5 base model in Figure 7. The POLYGLOSS quickly adapts to the new language, while the ByT5-based model never improves.

在图 7 中，当采用交错的 POLYGLOSS 模型和一个 ByT5 基础模型时，PORT MER 得分。POLYGLOSS 模型能够快速适应新语言，而基于 ByT5 的模型则从未有所提升。

9 Alignment Score as a Reward Function

9 对齐得分作为奖励函数

Our alignment score can also be used as a scalar reward function for reinforcement learning with verifiable rewards (RLVR, DeepSeek-AI and alia, 2025) when predicting a concatenated segmentation and glosses. We demonstrate this with a small pilot study using vanilla GRPO (Shao et al., 2024) to optimize the POLYGLOSS ByT5 Concatenated model on Gitksan. We use $\beta = 0.1$, $lr = 5E - 5$, batch and group sizes both 8, and train for 50 epochs. We use a temperature of 0.6, top-p of 0.9, and repetition penalty of 1.05 when sampling.

我们的对齐得分也可以用作可验证奖励 (RLVR, DeepSeek-AI 等, 2025) 的标量奖励函数，用于在预测连接的分段和词义时的强化学习。我们通过使用原始的 GRPO (Shao 等, 2024) 进行一个小规模的试点研究来展示这一点，以优化 POLYGLOSS ByT5 Concatenated 模型在 Gitksan 上的表现。我们使用 $\beta = 0.1$, $lr = 5E - 5$ ，批量和组大小均为 8，并训练 50 个 epoch。我们在采样时使用温度为 0.6，top-p 为 0.9，重复惩罚为 1.05。

We report results before and after tuning in Figure 8, observing that all three metrics improve slightly. This approach could be scaled to the full training dataset as an additional post-training step, providing a method to improve the model without any additional labeled data. Furthermore, RL could be used when adapting the model to a new language that lacks gold-labeled morphological seg-

我们在图 8 中报告了调优前后的结果，观察到所有三个指标都有小幅提升。这种方法可以作为额外的后训练步骤扩展到整个训练数据集上，从而提供一种在不使用任何额外标注数据的情况下改进模型的方法。此外，在将模型适应到缺乏黄金标注形态分割的新语言时，可以使用强化学习 (RL)。

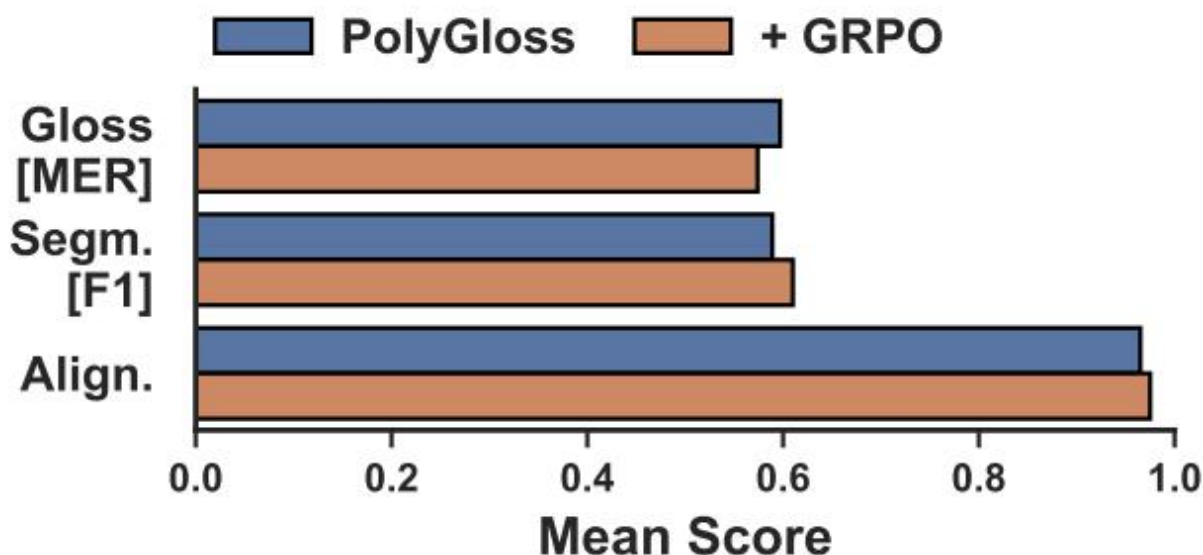


Figure 8: images/ef790ab900056cbf849a6ee65de94b6e566cd7af65ece6c3145427aef9e4d566.jpg

Figure 8: Scores before and after GRPO tuning for Gitksan.

图 8: Gitksan 在 GRPO 调优前后的得分。

mentations, since the alignment score will force the segmentation to (minimally) be aligned with the gold glosses.

由于对齐得分将强制分段与黄金释义（gold glosses）（至少）对齐。

10 Related Work

10 相关工作

Research has explored automatic interlinear glossing using a variety of techniques including active learning (Palmer et al., 2010, 2009), conditional random fields (Moeller and Hulden, 2018; McMillan-Major, 2020b), neural models (Moeller and Hulden, 2018; Zhao et al., 2020), and large language models (Ginn et al., 2024a; Yang et al., 2024; Elsner and Liu, 2025; Shandilya and Palmer, 2025). Our work is directly inspired by He et al. (2023), a submission to the 2023 SIGMORPHON Shared Task on Interlinear Glossing (Ginn et al., 2023), and Ginn et al. (2024b), both of which used multilingual pretraining on glossed text.

研究已经探讨了使用多种技术的自动线形注释，包括主动学习（Palmer 等，2010，2009）、条件随机场（Moeller 和 Hulden，2018；McMillan-Major，2020b）、神经模型（Moeller 和 Hulden，2018；Zhao 等，2020）以及大型语言模型（Ginn 等，2024a；Yang 等，2024；Elsner 和 Liu，2025；Shandilya 和 Palmer，2025）。我们的工作直接受到 He 等人（2023）的启发，该研究是提交给 2023 年 SIGMORPHON 线形注释共享任务（Ginn 等，2023）的，同时 Ginn 等人（2024b）也使用了对注释文本进行多语言预训练的方法。

Other work has focused on methods to automatically create IGT instances from other modalities, such as images of reference grammars (Round et al., 2020), LaTeX publications (Nordhoff and Krämer, 2022), and speech recordings (He et al., 2024). Recently, Aycock et al. and Yang et al. (2025b) proposed glossing as a method for testing LLMs’ abilities to apply grammatical knowledge.

其他工作则专注于从其他模态自动创建 IGT 实例的方法，例如参考语法规则的图像（Round 等，2020）、LaTeX 文献（Nordhoff 和 Krämer，2022）以及语音录音（He 等，2024）。最近，Aycock 等人 and Yang 等人（2025b）提出使用“注释”作为一种测试大型语言模型（LLMs）应用语法知识能力的方法。

11 Conclusion

11 结论

Tools for language documentation are only valuable when designed with the user—annotators and linguists—in mind. We address user feedback on automated glossing models and develop new multilingual models with major improvements over prior work. Most significantly, the POLYGLOSS models predict both morphological segments and interlinear glosses in a single forward pass, enabling more useful, interpretable, and trustworthy suggestions. We set a new state-of-the-art for glossing in several languages, as well as optimizing our model for segmentation accuracy and alignment between the two tasks. Finally, we offer practical recommendations for predicting performance and adapting the model to new data, keeping in mind computational constraints.

语言文档工具只有在设计时考虑到用户（标注员和语言学家）的需求时才具有价值。我们针对自动词性标注模型的用户反馈进行了分析，并开发了新的多语言模型，显著改进了以往的工作。最重要的是，POLYGLOSS 模型能够在一次前向传递中预测形态学片段和双行注释，从而提供更有用、更易解释且更可信的建议。我们在几种语言的词性标注方面设定了新的最先进水平，并优化了模型在分段准确性和两项任务之间对齐方面的表现。最后，我们提供了实用的建议，用于预测性能并适应新数据，同时考虑到计算限制。

Going forward, we plan to work with developers of annotation software such as ELAN (The Language Archive, 2025) and FLEX (SIL Global, 2026) to integrate our model into real-world documentation workflows. One straightforward solution is to use our model to predict segmentation and gloss lines for samples with transcribed text, filter predictions according to model confidence, and allow the user to accept, reject, or edit the predictions.

今后，我们计划与诸如 ELAN (The Language Archive, 2025) 和 FLEX (SIL Global, 2026) 等标注软件的开发者合作，将我们的模型整合到实际的文档工作中。一个直接的解决方案是使用我们的模型来预测带有转录文本的样本的分段和术语线，根据模型的置信度过滤预测结果，并允许用户接受、拒绝或编辑这些预测结果。

Limitations

限制

We do not compare against closed-sourced LLMs for several reasons. First, the training datasets for these models are opaque, and with much of the test data being available online, there is risk of contamination (of course, this may also be true for the Qwen and Cohere models as well). Second, endangered language data often bears considerations of data sovereignty, and language communities often do not want their data to be sent to a third-party provider. Our models are open-source and open-weights and can be finetuned locally.

我们不会将模型与闭源的 LLM 进行比较，原因有几个。首先，这些模型的训练数据集是不透明的，而大部分测试数据在线上可以获取，存在数据污染的风险（当然，这可能也适用于 Qwen 和 Cohere 模型）。其次，濒危语言数据通常涉及数据主权的考虑，语言社区通常不希望他们的数据被发送给第三方提供商。我们的模型是开源且开放权重的，可以在本地进行微调。

For morphological segmentation, we do not differentiate between surface-level morphemes and underlying-form morphemes (and our dataset includes both). Thus, the task is not exactly identical across datasets, and there is no guarantee that the predicted segmentation may exactly match the input string.

对于形态学分词，我们不区分表层形态素和底层形态素（我们的数据集包含这两种形态素）。因此，不同数据集之间的任务并不完全相同，无法保证预测的分词结果与输入字符串完全一致。

We do not attempt to standardize the glosses in our dataset for two reasons. First, conventions and meanings vary greatly across annotators, and there is no way we could ensure the original intent of the annotator was preserved for the thousands of examples and languages in our dataset. Second, Ginn et al. (2024b) tried standardizing glosses to the UniMorph conventions (Batsuren et al., 2022) and found no evidence that it improved performance. Instead, our model outputs glosses according to the various conventions in its training data, and it can be adapted to a new convention via low-rank finetuning.

我们没有尝试在数据集中对注释的术语进行标准化，原因有两个。首先，不同注释者之间的惯例和含义差异很大，我们无法确保在数据集中数千个例子和语言中保留注释者的原始意图。其次，Ginn 等人 (2024b) 尝试将术语标准化为 UniMorph 的惯例 (Batsuren 等人, 2022)，但并未发现这能提高性能。相反，我们的模型会根据其训练数据中的各种惯例输出术语，并且可以通过低秩微调适应新的惯例。

Ethical Considerations

伦理考量

Automatic approaches to interlinear glossing are intended to help accelerate the language documentation process and contribute to the fight against language death. However, there is risk of misuse, and these systems should not fully replace human annotators, which could result in er-

aneous documentation that hinders downstream applications. All data was taken from existing work and used in accordance with the original stakeholders' wishes. Finally, our work used a large amount of computational resources, which inevitably bears an environmental cost.

自动标注方法旨在帮助加快语言记录过程，并有助于对抗语言消亡。然而，这些方法存在误用的风险，这些系统不应完全取代人工标注员，否则可能导致错误的记录，从而阻碍后续应用。所有数据均来自已有研究，并按照原始利益相关者的意愿进行使用。最后，我们的工作使用了大量计算资源，这不可避免地带来了环境成本。

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任何意见、发现、结论或建议均代表作者的观点，不一定反映国家自然科学基金会的看法。

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